

Math 110 – Homework 10 Answers

Math 110 – Fall 2026

Homework 10

Orthogonality, Gram–Schmidt, and projections

Suggested completion date: Friday, November 6

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. On $\mathcal{P}_2(\mathbb{R})$ use the inner product

$$\langle p, q \rangle = \int_{-1}^1 p(x)q(x) dx.$$

Apply the Gram–Schmidt procedure to the ordered basis

$$1, x, x^2$$

to obtain an orthonormal basis of $\mathcal{P}_2(\mathbb{R})$.

Problem 2. Let

$$W = \text{span}((1, -1, 1, 0), (0, 2, 1, 0)) \subseteq \mathbb{R}^4$$

with the standard inner product. Find a basis of $W^\perp$.

Problem 3. On $\mathcal{P}_2(\mathbb{R})$ use the same inner product as in Problem 1, and let

$$W = \text{span}(x + 1).$$

Find a basis of $W^\perp$.

Problem 4. Let U and W be subspaces of a finite-dimensional inner-product space V . Prove

$$(U \cap W)^\perp = U^\perp + W^\perp.$$

Problem 5. Let U be a subspace of a finite-dimensional inner-product space V , and let $e_1, \dots, e_m$ be an orthonormal basis of U .

(a) Prove that

$$P_U v = \sum_{j=1}^m \langle v, e_j \rangle e_j.$$

(b) Prove the best-approximation property

$$\|v - P_U v\| \leq \|v - u\| \quad \text{for every } u \in U,$$

and determine when equality holds.

Problem 6 (optional). On $\mathcal{P}_1(\mathbb{R})$ use the inner product

$$\langle f, g \rangle = \int_{-1}^1 f(x)g(x) dx.$$

Find the unique polynomial $p \in \mathcal{P}_1(\mathbb{R})$ such that

$$\int_0^2 f(x) dx = \langle f, p \rangle$$

for every $f \in \mathcal{P}_1(\mathbb{R})$.

Problem 7 (optional). Let P be a projector on a finite-dimensional inner-product space, so $P^2 = P$. Prove that P is an orthogonal projection if and only if

$$\|Pv\| \leq \|v\| \quad \text{for every } v.$$

Solutions

Problem 1

Solution notes

Problem 2

Solution notes

Problem 3

Solution notes

Problem 4

Solution notes

Problem 5**Solution notes**

Problem 6**Solution notes**

Problem 7**Solution notes**
